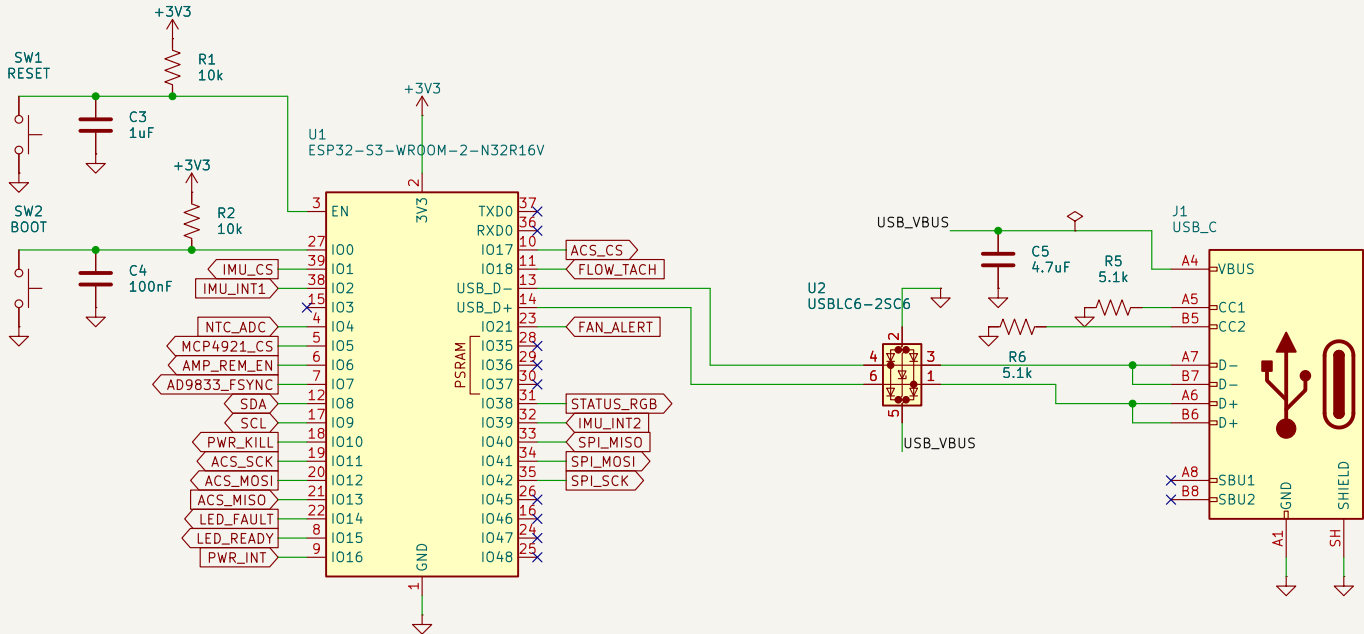
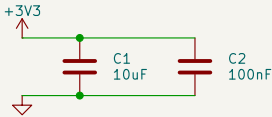


ESP32-S3-WROOM-2-N32R16V: 32 MB octal flash + 16 MB octal PSRAM (GPIO26-37 reserved by module)

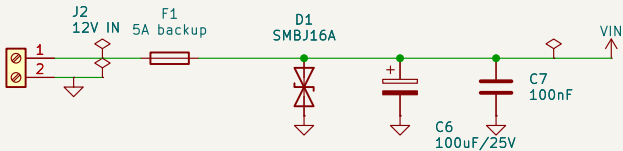
GPIO map follows include/config/pin_config.h

ESP32-S3 DECOUPLING CAPACITORS



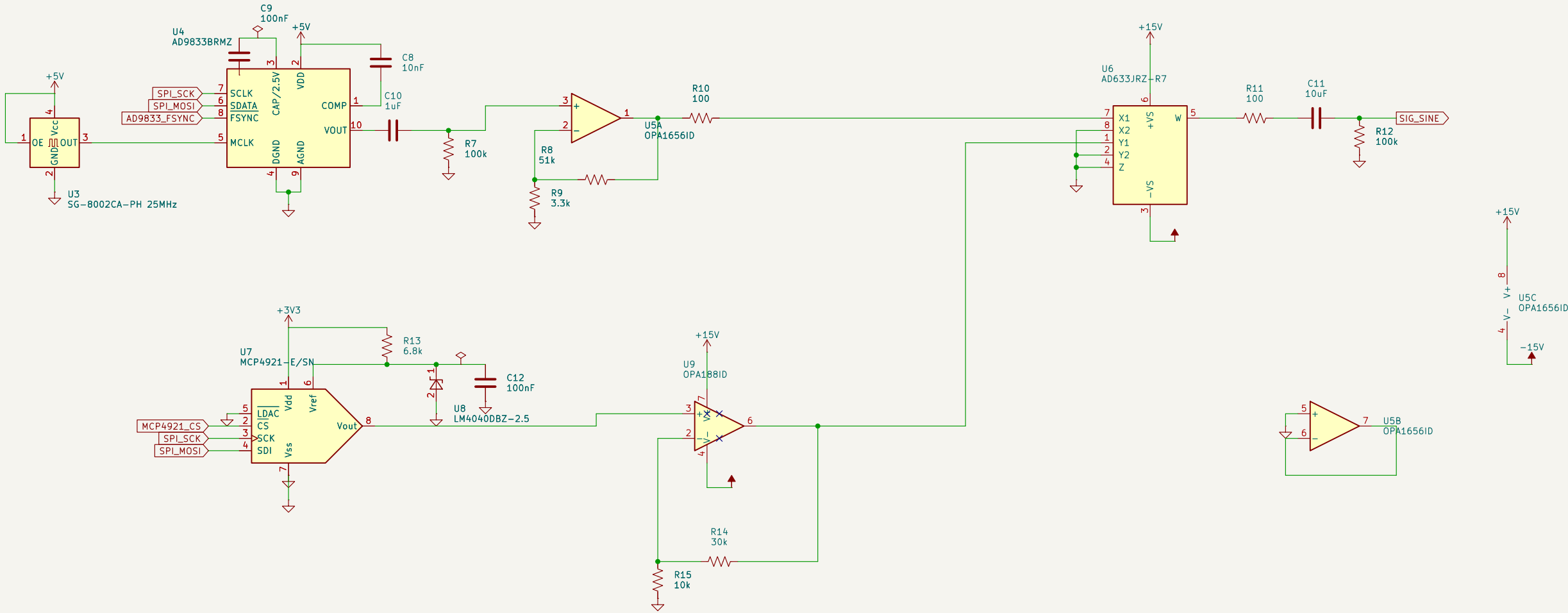
12 V input from a Dell XYK93 (DPS–2000EB class, ~12.2 V, ~160 A available): use the main 12 V rail with PS_ON held low.

F1 is a backup only: add an inline high–breaking–capacity fuse (7.5–10 A) at the PSU end. Reverse polarity: TVS forward–clamps and blows the fuse.

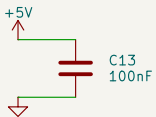


AD9833 60 Hz sine -> OPA1656 gain stage -> AD633 multiplier; MCP4921 + OPA188 give a 0-10 V amplitude control.

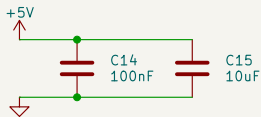
Vctl 0-10 V (2.5 V ref x4) x sine +/-4.9 V pk / 10 = 0..4.9 V pk (9.9 Vpp, 3.5 Vrms) to amplifier input; ZE500.1 RCA input range 0.2-6 V.



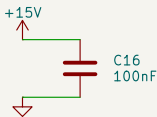
SG-8002CA DECOUPLING CAPACITORS



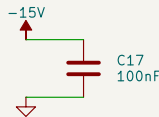
AD9833 DECOUPLING CAPACITORS



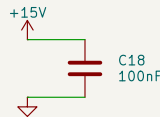
OPA1656 +15V DECOUPLING CAPACITORS



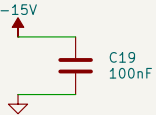
OPA1656 -15V DECOUPLING CAPACITORS



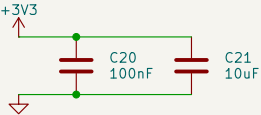
AD633 +15V DECOUPLING CAPACITORS



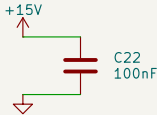
AD633 -15V DECOUPLING CAPACITORS



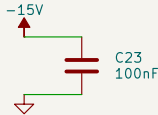
MCP4921 DECOUPLING CAPACITORS



OPA188 +15V DECOUPLING CAPACITORS



OPA188 -15V DECOUPLING CAPACITORS



Cryocooler Controller

Sheet: /Signal Generator/
File: signal_gen.kicad_sch

Title: Signal Generator

Size: A3

Date:

Rev: A

KiCad E.D.A. 10.0.6

Id: 3/13

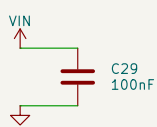
EMC2303: ADDR_SEL 33k pull-up -> SMBus 0x4D (datasheet Table 5-1); CLK 4.7k pull-up -> default fan drive 0%. ch1 fans, ch2 pump. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.

EMC2303 DECOUPLING CAPACITORS

Size: A3 Date: Rev: A
 Sheet: /Cooling/ File: cooling.kicad_sch
 Title: Cooling
 KICad E.D.A. 10.0.6 Id: 4/13

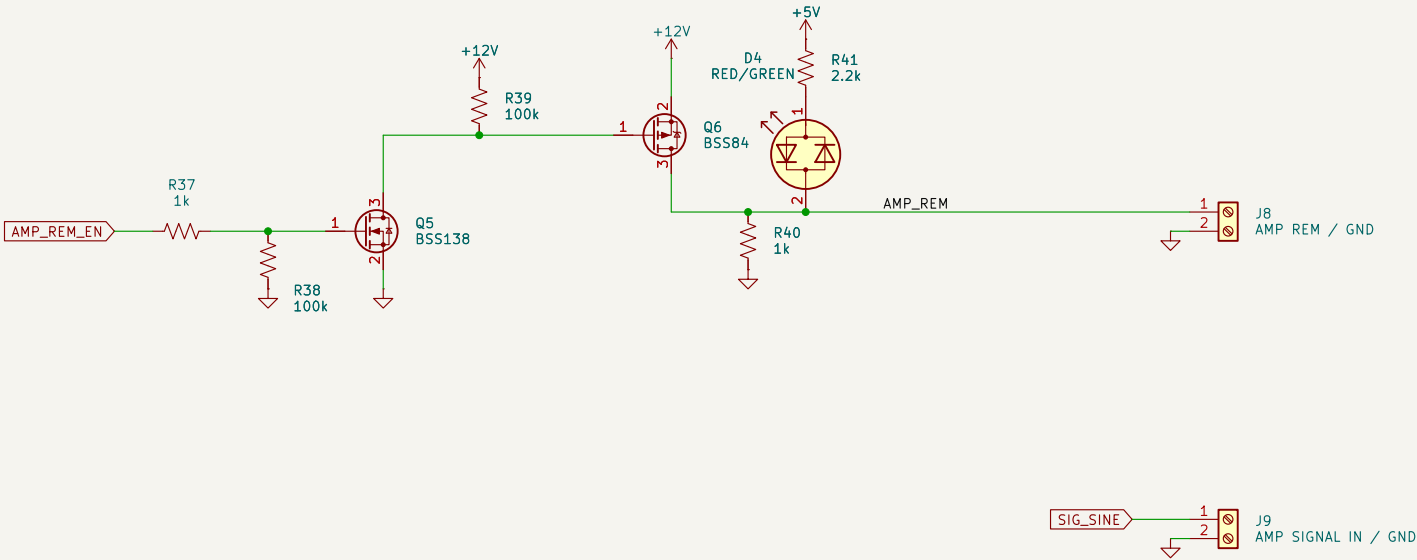


PWR_ON follows the +3V3 rail: on = green, standby = red. CL25 (25 mA) feeds both legs, ~9 mA reaches the LED, ~25 mA total from VIN in standby



GPIO6 high -> +12 V on the amplifier REM terminal (P-FET high-side switch). LED: green = on, red = off.

Amplifier: AUDIOZERONE ZE500.1 (external): REM terminal + RCA/line input. Its power leads do not pass through this board.



Cryocooler Controller

Sheet: /Amp Control/
File: amp_control.kicad_sch

Title: Amp Control

Size: A3

Date:

Rev: A

KiCad E.D.A. 10.0.6

Id: 6/13

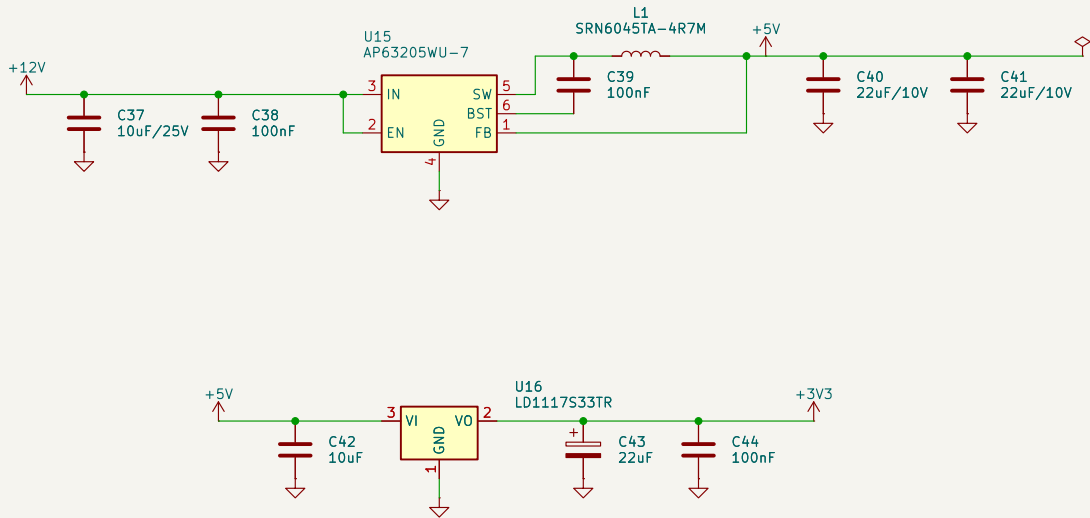
PT1000 4-wire ratiometric: IDAC1 (AIN3) excites, AIN1/AIN0 sense, Ref (3.9k) on REFP/REFN. I2C 0x45 (A0=A1=DVDD).

ADS122C04 DECOUPLING CAPACITORS

Cryocooler Controller	
Sheet: /Cold Head/ File: cold_head.kicad_sch	
Title: Cold Head	
Size: A3	Date:
KiCad E.D.A. 10.0.6	Rev: A Id: 7/13

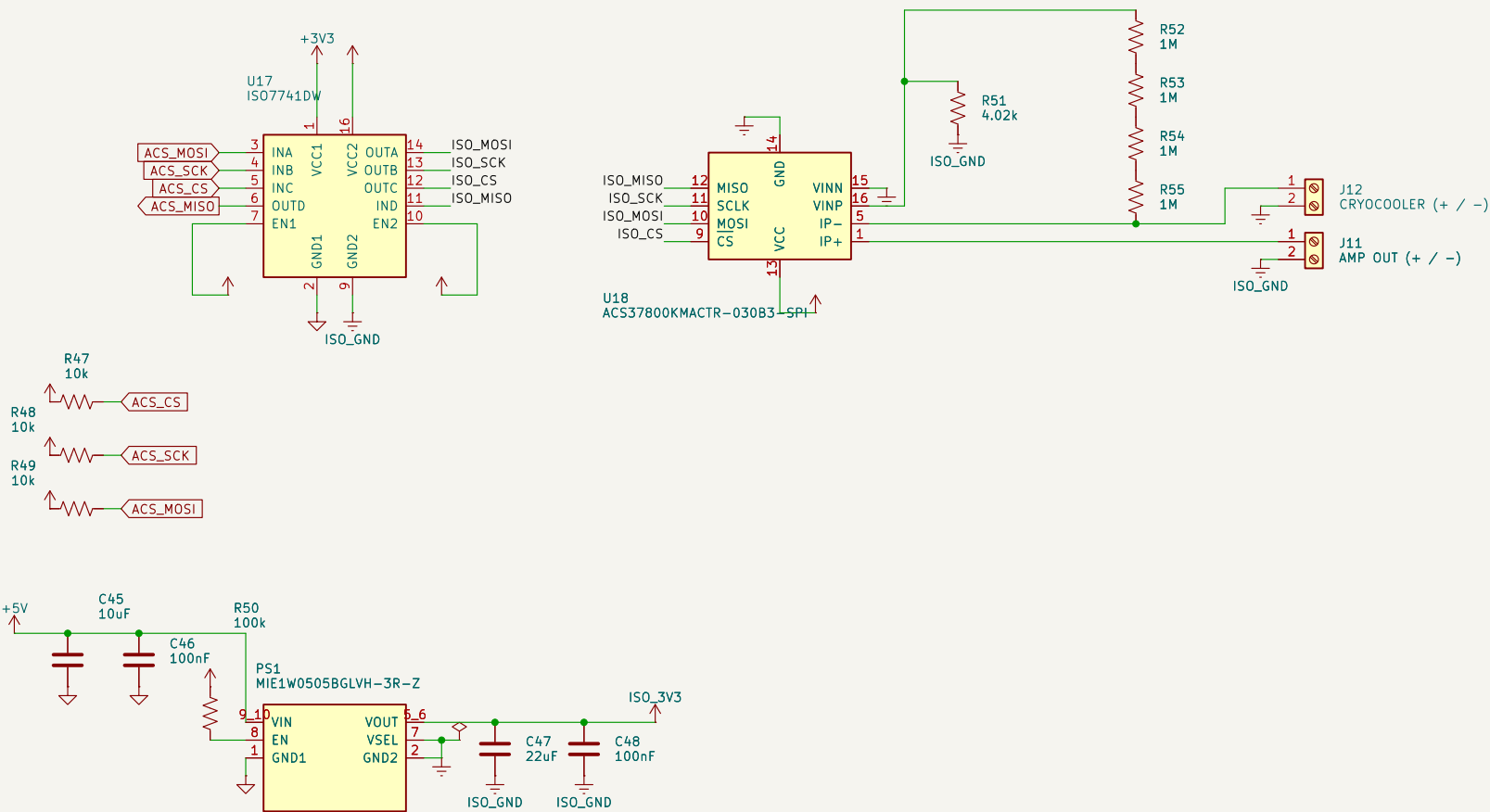
12 V -> 5 V (AP63205 buck, datasheet Fig. 1 / Table 3: 4.7 uH, 10 uF in, 2 x 22 uF out, 100 nF bootstrap) -> 3.3 V (LD1117S33).

AP63205: VIN 3.8–32 V (40 V for 400 ms), EN tied to VIN, FB tied to the fixed 5 V output.

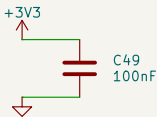


Amplifier output V/I monitor: ACS37800 (SPI) on an isolated 3.3 V domain (MIE1W0505, VSEL=GND2 -> 3.3 V); SPI bus GPIO11/12/13, CS GPIO17.

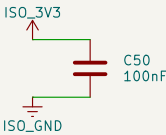
Mains-class creepage/clearance applies between ISO_GND domain and the logic side on the PCB.



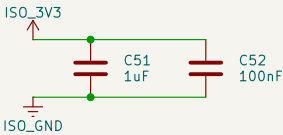
ISO7741 VCC1 DECOUPLING CAPACITORS



ISO7741 VCC2 DECOUPLING CAPACITORS



ACS37800 DECOUPLING CAPACITORS



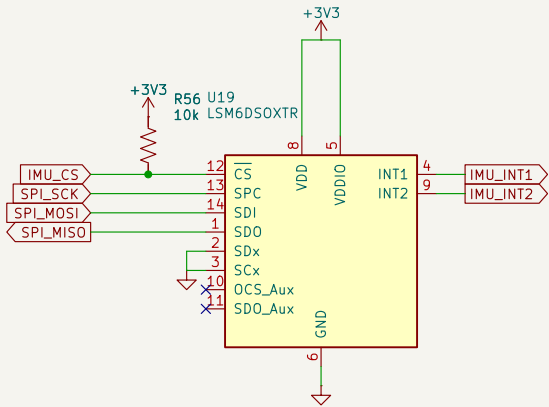
Cryocooler Controller

Sheet: /Amp Current/
File: amp_current.kicad_sch

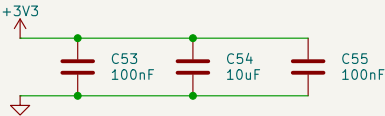
Title: Amp Current

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 9/13

LSM6DSOX in 4-wire SPI mode on the main bus (CS GPIO1, INT1 GPIO2 single-tap, INT2 GPIO39 wake-up).



LSM6DSOX DECOUPLING CAPACITORS



Cryocooler Controller

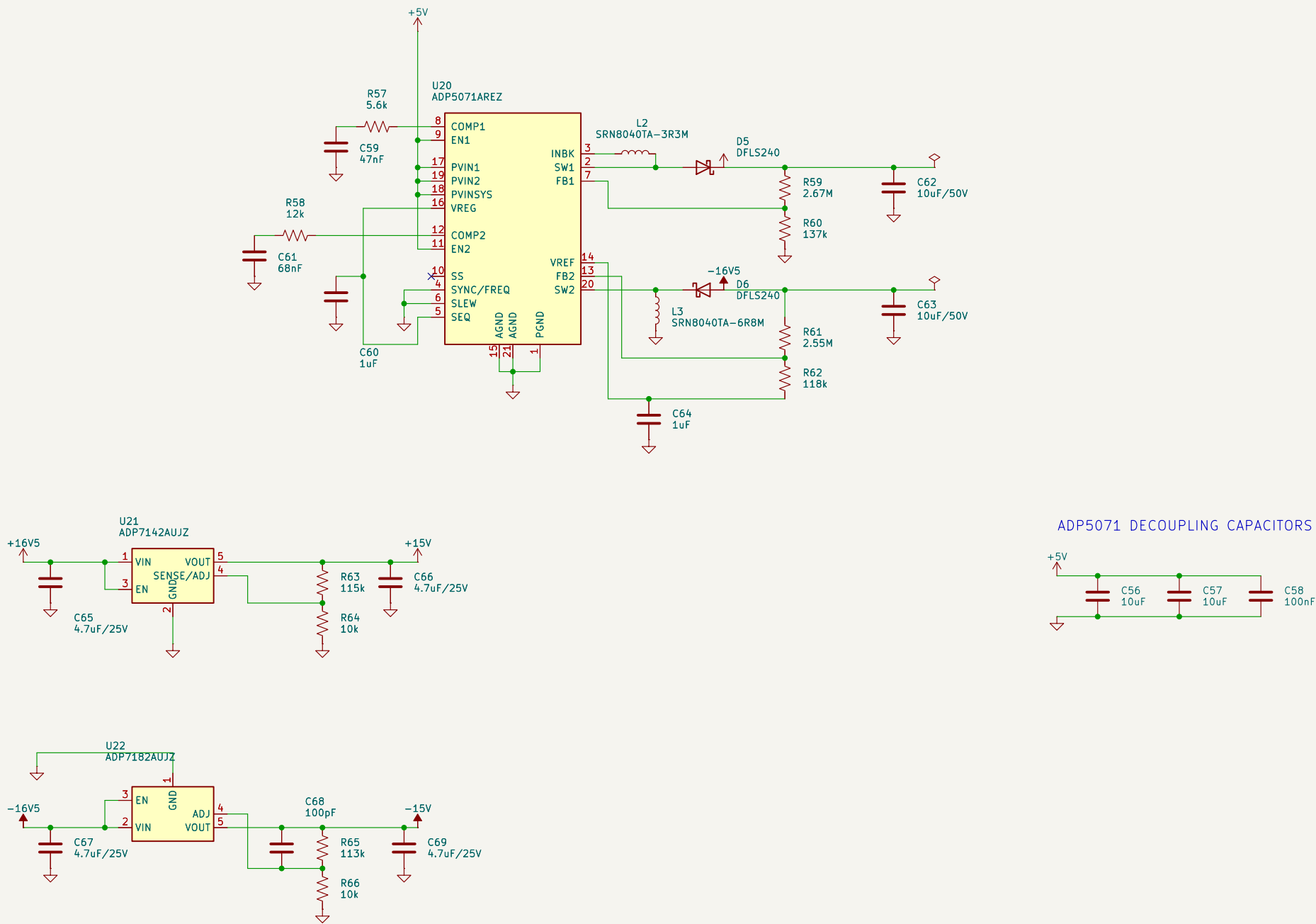
Sheet: /IMU/
File: imu.kicad_sch

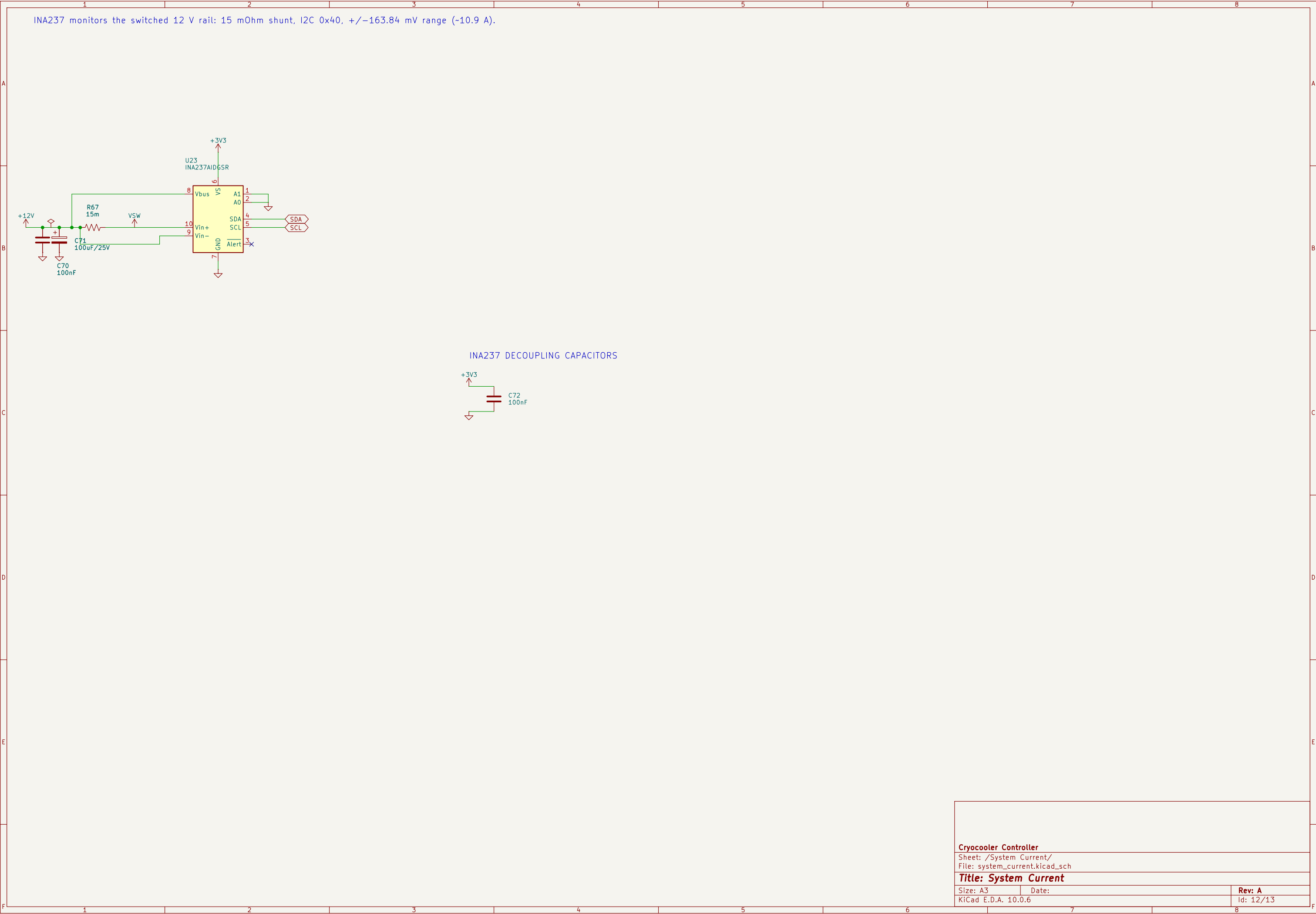
Title: IMU

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 10/13

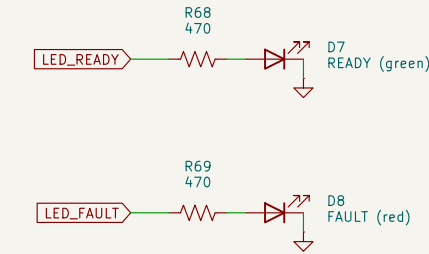
ADP5071AREZ (5 V in): +16.5 V boost and −16.5 V inverting rails; ADP7142 (+15 V) and ADP7182 (−15 V) post-regulate.

Values from ADP5071 datasheet Fig. 47 / Tables 9–11 (+/−16.5 V output rows rescaled via RFT). ADP7182 EN tied to VIN (auto-start); |VEN| ≥ 2 V enables.

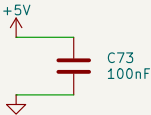




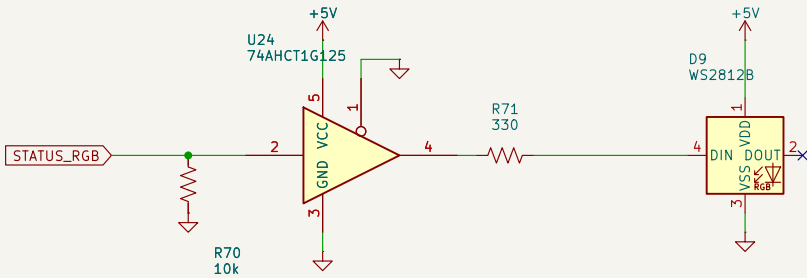
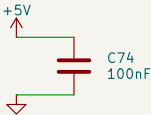
READY (GPIO15) and FAULT (GPIO14) LEDs, active high; WS2812B status LED on GPIO38 via a 74AHCT1G125 3.3 V -> 5 V level shifter (WS2812B VIH = 0.7 x VDD).



74AHCT1G125 DECOUPLING CAPACITORS



WS2812B DECOUPLING CAPACITORS



Cryocooler Controller

Sheet: /Indicators/
File: indicators.kicad_sch

Title: Indicators

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.6		Id: 13/13